On the Design and Validation of the MISSat-1 Small Satellite Ground Station

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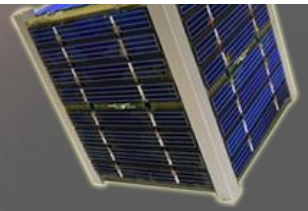

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MISSat-1

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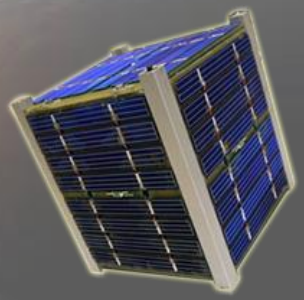
Introduction

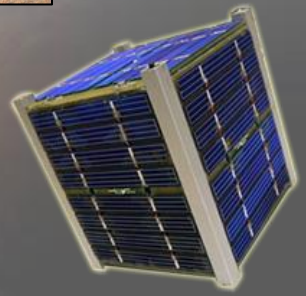
- Purpose of the ground station is to communicate with small satellites.
- Main Functions:
  - Track orbiting small satellites, especially MISSat-1
  - To download telemetry data and pictures taken by MISSat-1

Ground Station Antennas



Components

- UHF and VHF antennas
- Preamplifiers
- Rotator
- Roof-Top tower
- Coax
- Transceiver
- Computer
- Misc. Parts (cables, antenna supplies, etc.)



Functionality

- The satellite tracking software, Ham Radio Deluxe (HRD), is used to provide radio and rotator control information to the computer.
- The computer uses an Arduino microcontroller to remotely control the rotator's power controller.
- The rotator will change the azimuth and elevation of the mounted antennas to track a satellite as it travels across the sky.

Functionality

- HRD provides computer control of the radio while tracking by setting the radio's frequency to compensate for Doppler shift.

Components: UHF Antenna

- 436CP30 Circular Polarized Yagi antenna from M2 Antenna Systems, Inc. [1]
- 432-440 MHz (70cm band)
- Gain: 15.50 dBic
- Beamwidth: 30 degrees circular
- Cost: $291.00
- PS-70CM R/L Switch
  - $177.00

Components: VHF Antenna

- 2MCP14 Circular Polarized Yagi antenna from M2 Antenna Systems, Inc. [1]
- 143-148 MHz (2m band)
- Gain: 12.34 dBic
- Beamwidth: 30 degrees circular
- Cost: $266.00
- PS-2M R/L Switch
  - $177.00

Components: Preamplifiers

- SP-2000 (144 MHz) and SP-7000 (440 MHz)
- Gain: 10-20 dB
- Powered by common 12V DC source
- Cost:
  - SP-2000: $330.00
  - SP-7000: $330.00

Components: Preamplifiers

Preamplifiers used by the University of Mississippi Ground Station.

Preamplifier DC Power Source

Components: Rotator

- Yaesu G-5500 (AZ-EL)
- Cost: $699.95

Photo credit: texastowers.com

Components: Rotator

Rotator used by the University of Mississippi Ground Station.

Components: Roof-Top tower

- 8’ Roof-Top Tower (RT-832) from Glen Martin Engineering
- 2 Mast Clamps (MC-10)
- 10’ HD Aluminum Mast (MA-2010)
- Total Cost: $622.00

Components: Coax

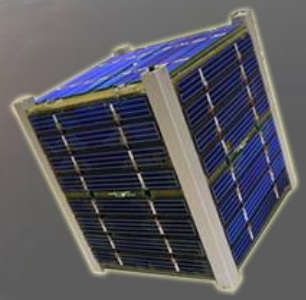
- CNT-600 (N-PL and N-N)
- Attenuation dB/100ft
  - 150 MHz: 0.97 dB
  - 450 MHz: 1.71 dB
- Cost per 100ft:
  - N-PL $187.95
  - N-N $168.95

Photo credit: commscope.com

Components: Transceiver

- Kenwood TS 2000 HF/VHF/UHF transceiver
- Output power:
  - HF, 6m, 2m: 100 watts
  - UHF: 50 watts
- Cost: $1474.99

Transceiver used by the University of Mississippi ground station.



III. Components: Computer

- Intel Core 2 Duo 1.86 GHz Processor
- 6 GB RAM
- HRD Satellite Tracking Software
- Arduino Duemilanove rotator controller serial interface

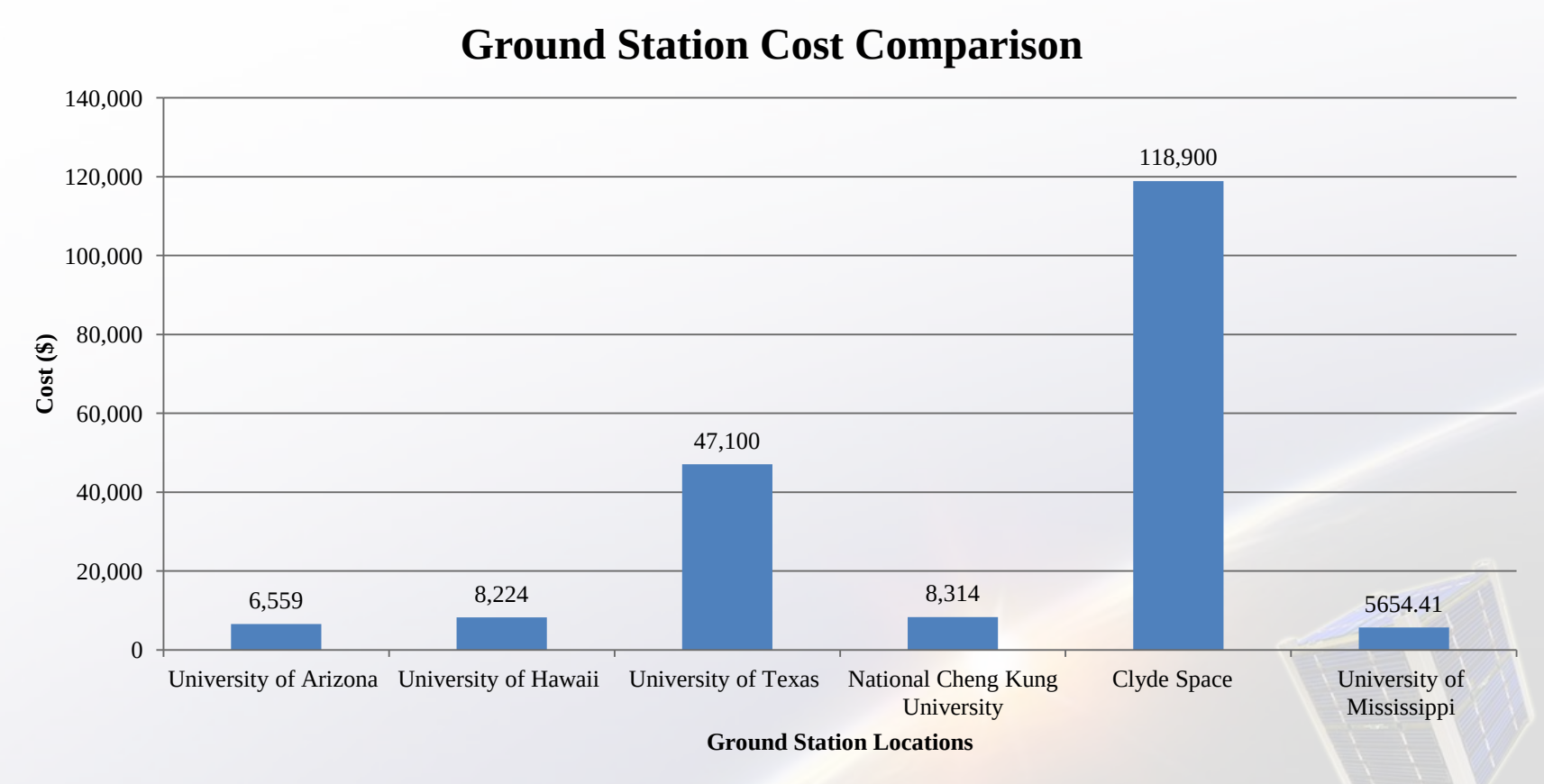
UM Ground Station Cost Breakdown

ITEM	MODEL	USD
UHF Antenna with R/L Switch	2MCP22	$443.00
VHF Antenna with R/L Switch	436CP30	$468.00
Preamplifiers	SP-2000/7000	$330.00/$330.00
Rotator	Yaesu G-5500	$699.95
Roof-Top tower	RT-832	$622.00
Coax	CNT-600/240	$356.90/$43.20
Radio	Kenwood TS-2000	$1474.99
Radio PSU	Kenwood KPS-15	$139.99
Misc Parts	Cables, antenna supplies, etc	$746.38
Total Ground Station Cost		**$5654.41**

Cost Comparison

	Antenna ($)	Transceiver ($)	Preamplifiers ($)	Coax & connectors ($)	Rotator ($)
University of Arizona [2]	720.00	1,450.00	1203.00	374.95	260.00
National Cheng Kung University [3]	575.00	3030.00	263.00	700.00	1,060.00
University of Hawaii [4]	358.00	870.00	330.00	20.00	639.95
University of Texas [5]	20,000.00	1,500.00	275.00	4,500.00	
Clyde Space [6]	1,033.80				1,239.95
University of Mississippi	911.00	1474.99	660.00	400.10	699.95

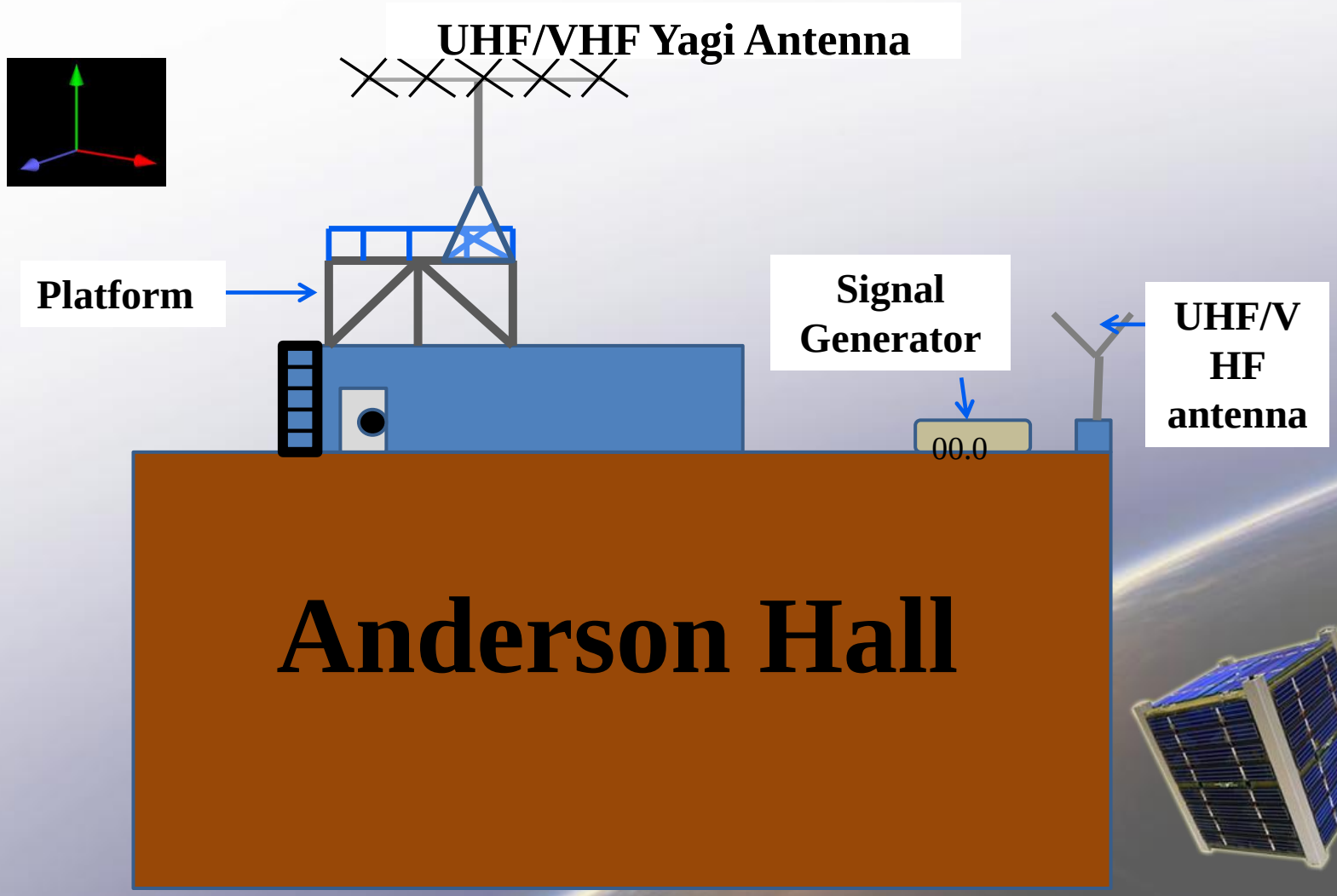
Cost Comparison

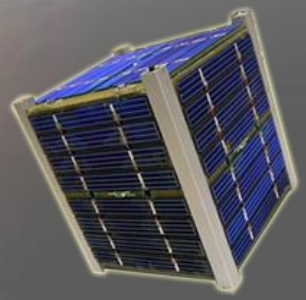


Programming and Setup of Instrumentation

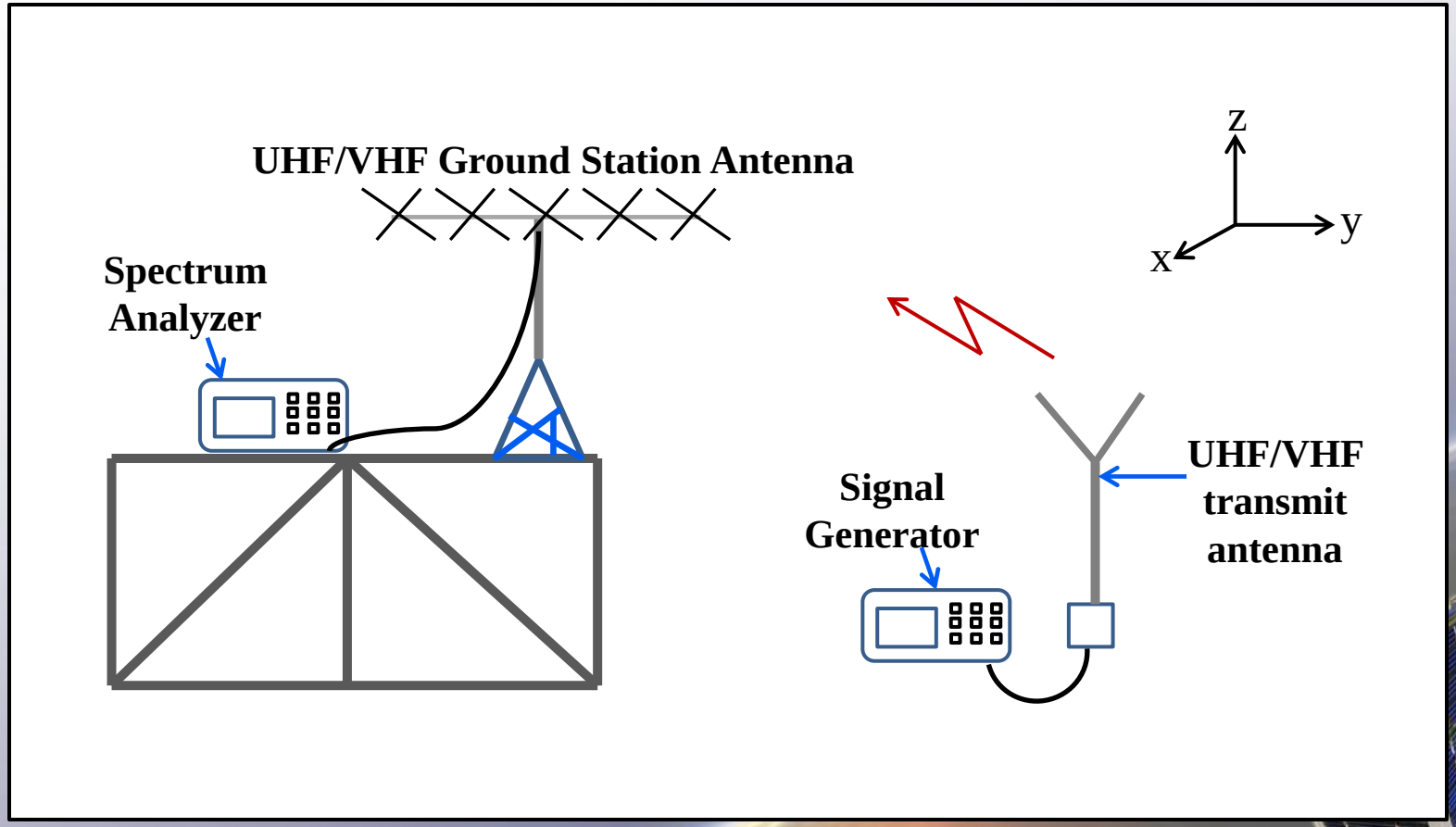
- A program was written in MATLAB that would control rotator and spectrum analyzer functions. Azimuth angle and amplitude levels are recorded by the program as the rotator turns the antennas.
- Outdoor setup involved a signal generator and small UHF/VHF antenna about 100 feet away from ground station antennas.

Outdoor Measurement Setup

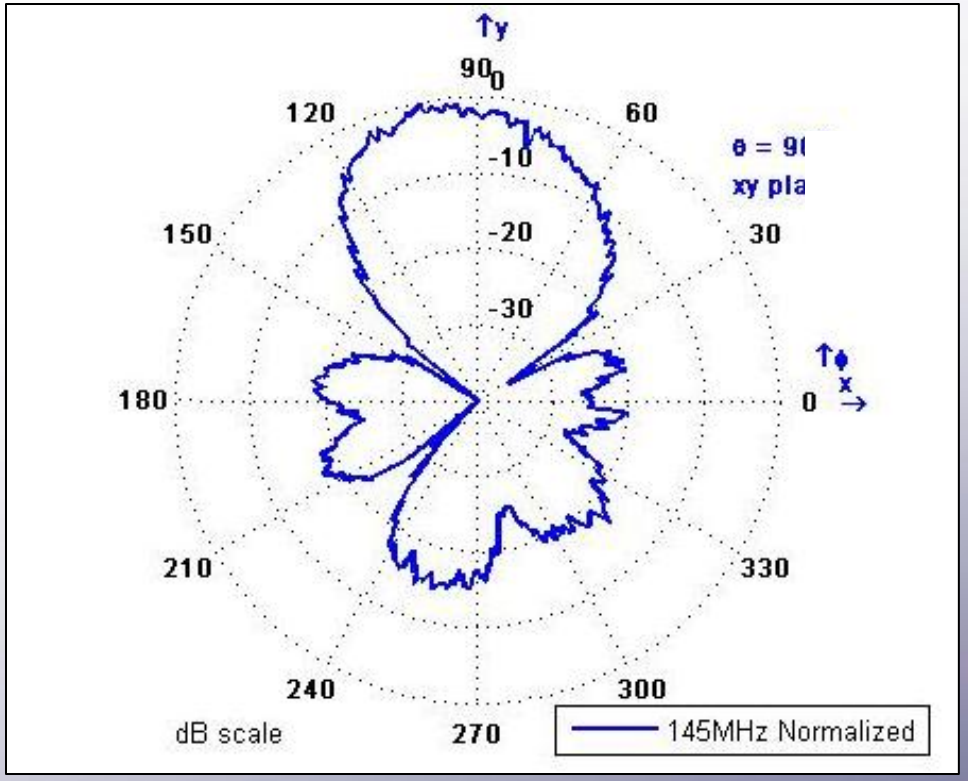


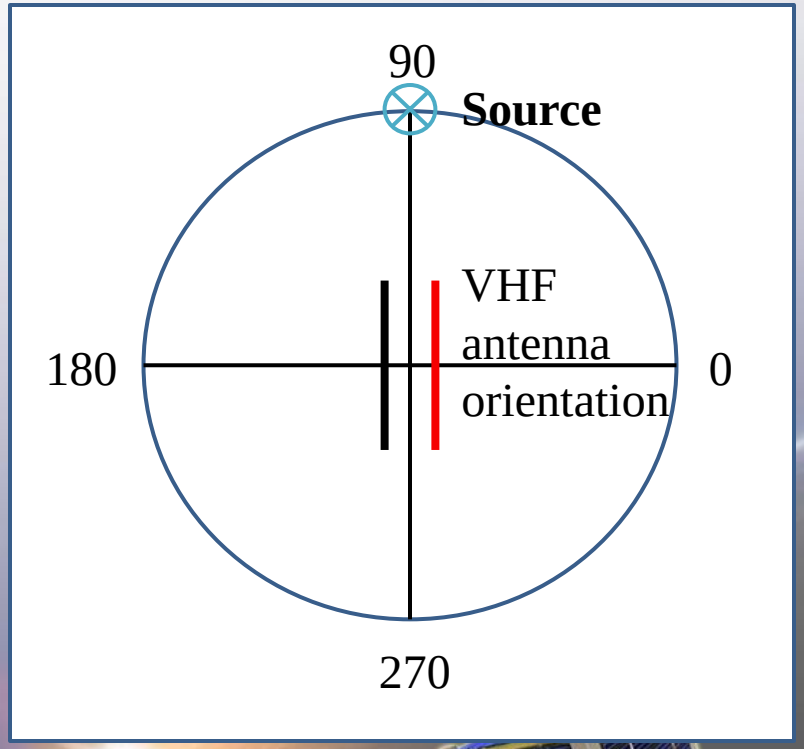


Logical Component Arrangement



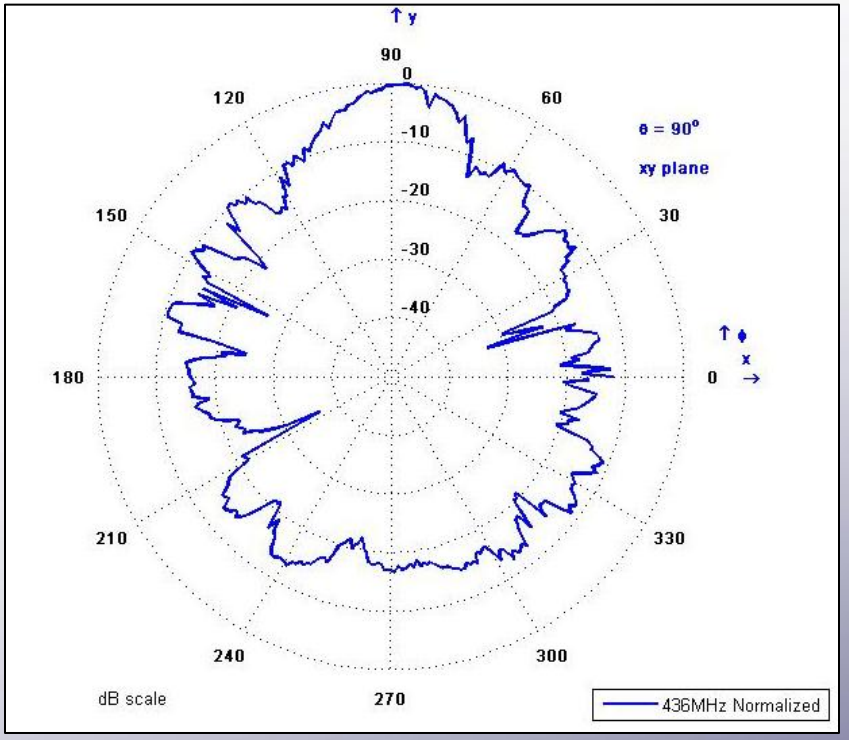
Normalized Radiation Pattern of VHF Antenna at 145 MHz

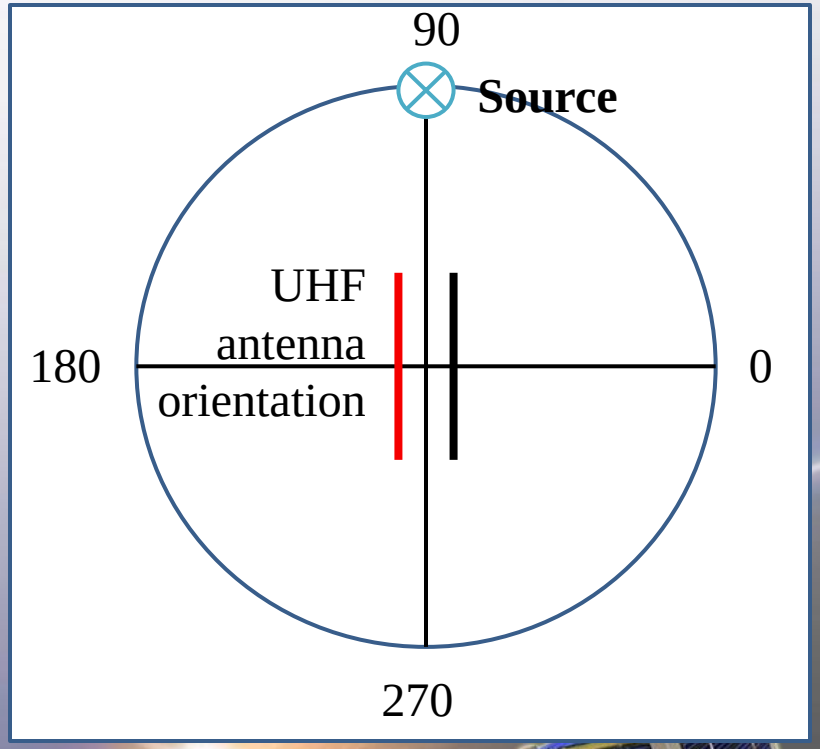




HPBW = 33º; Side lobes: below -15dB; Front to back ratio: ~15dB

Normalized radiation pattern of UHF antenna at 436 MHz



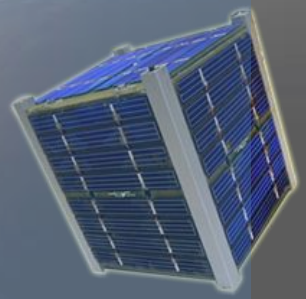


HPBW = 22°; Side lobes: below -9 dB; Front to back ratio: ~ 12 dB

Uplink Calculations

$$P_R = P_T - L_{FS} - L_{ATM} - L_{POL} - L_{POINT} - L_{FEED} + G_T + G_R - 3dB + 17dB$$

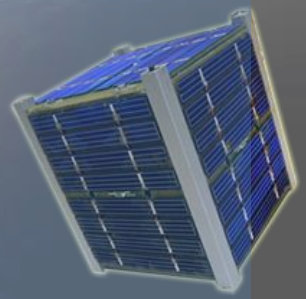
$P_R \rightarrow$ Power Received by Satellite, dB_W
$P_T \rightarrow$ Power Transmitted by Ground station, dB_W
$L_{FS} \rightarrow$ Free Space Loss, dB
$L_{ATM} \rightarrow$ Atmospheric Loss, dB
$L_{POL} \rightarrow$ Polarization Loss, dB
$L_{POINT} \rightarrow$ Pointing Loss, dB
$L_{FEED} \rightarrow$ Feeder Loss, dB
$G_T \rightarrow$ Ground station Antenna Gain, dB
$G_R \rightarrow$ Satellite Antenna Gain, dB
$3\ dB \rightarrow$ Gain Margin, dB
$17\ dB \rightarrow$ Preamplifier Gain, dB



Downlink Calculations

$$P_R = P_T - L_{FS} - L_{ATM} - L_{POL} - L_{POINT} - L_{FEED} + G_T + G_R - 3dB + 17dB$$

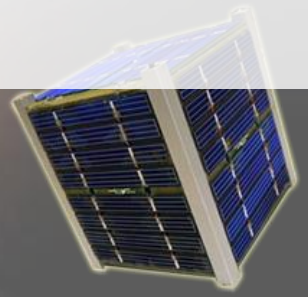
$P_R \rightarrow$ Power Received by Ground station, dB_W
$P_T \rightarrow$ Power Transmitted by Satellite, dB_W
$L_{FS} \rightarrow$ Free Space Loss, dB
$L_{ATM} \rightarrow$ Atmospheric Loss, dB
$L_{POL} \rightarrow$ Polarization Loss, dB
$L_{POINT} \rightarrow$ Pointing Loss, dB
$L_{FEED} \rightarrow$ Feeder Loss, dB
$G_T \rightarrow$ Satellite Antenna Gain, dB
$G_R \rightarrow$ Ground station Antenna Gain, dB
$3\ dB \rightarrow$ Gain Margin, dB
$17\ dB \rightarrow$ Preamplifier Gain, dB



Link Budget Summary

	P_T (dB_W)	L_{FS} (dB)	L_{ATM} (dB)	L_{POL} (dB)	L_{POINT} (dB)	L_{FEED} (dB)	G_T (dB)	G_R (dB)	P_R (dB_W)
Uplink	20	144.7755	3	3	2	4.5	16	0	-107.2755
Downlink	1.25	154.0563	3	3	2	4.5	0	16	-135.3063

- As seen above, the downlink power received is slightly higher due to a higher transmit power and higher free space loss



Successful Contacts

We have contacted the following amateur radio satellites UHF/VHF FM repeaters.

- AO-7 Amsat-Oscar 7
- SO-50

VI. Conclusions

- Due to alignment issues between the ground station and source antennas, results differed from manufacturer specifications.

VI. Conclusions

- The first test of the University of Mississippi ground station yields an adequate performance to communicate with small satellites in low earth orbit. Effective communication with small satellites can be achieved with a low budget ground station.

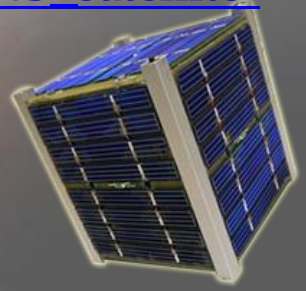
VII. References

[1] m2inc.com

[2] ftp://pirlftp.lpl.arizona.edu/cubesat/cubesat_papers/gspaper.pdf

[3] http://citeseerx.ist.psu.edu/

[4] http://www.spacegrant.hawaii.edu/reports/16_SUM06 FA06/Ichikawa_Dylan_FA06.pdf

[5] http://courses.ae.utexas.edu/ase463q/design_pages/spring03/ satellite_tracking/contents.htm

[6] http://www.clyde-space.com/cubesat_shop/ground_stations/343_satellite-tracking-and-control-station-stac